Bones

The Support System and More

Skeletal Cartilage

- Contains no blood vessels or nerves
- Surrounded by the perichondrium (dense irregular connective tissue) that resists outward expansion
- Three types
  - Hyaline
  - Elastic
  - fibrocartilage

Hyaline Cartilage

- the most abundant skeletal cartilage
- Support, flexibility, and resilience

- Present in these cartilages:
  - Articular – covers the ends of long bones
  - Costal – connects the ribs to the sternum
  - Respiratory – makes up the larynx and reinforces air passages
  - Nasal – supports the nose

Elastic Cartilage

- Similar to hyaline cartilage but contains elastic fibers
- Found in:
  - the external ear
  - the epiglottis

Fibrocartilage

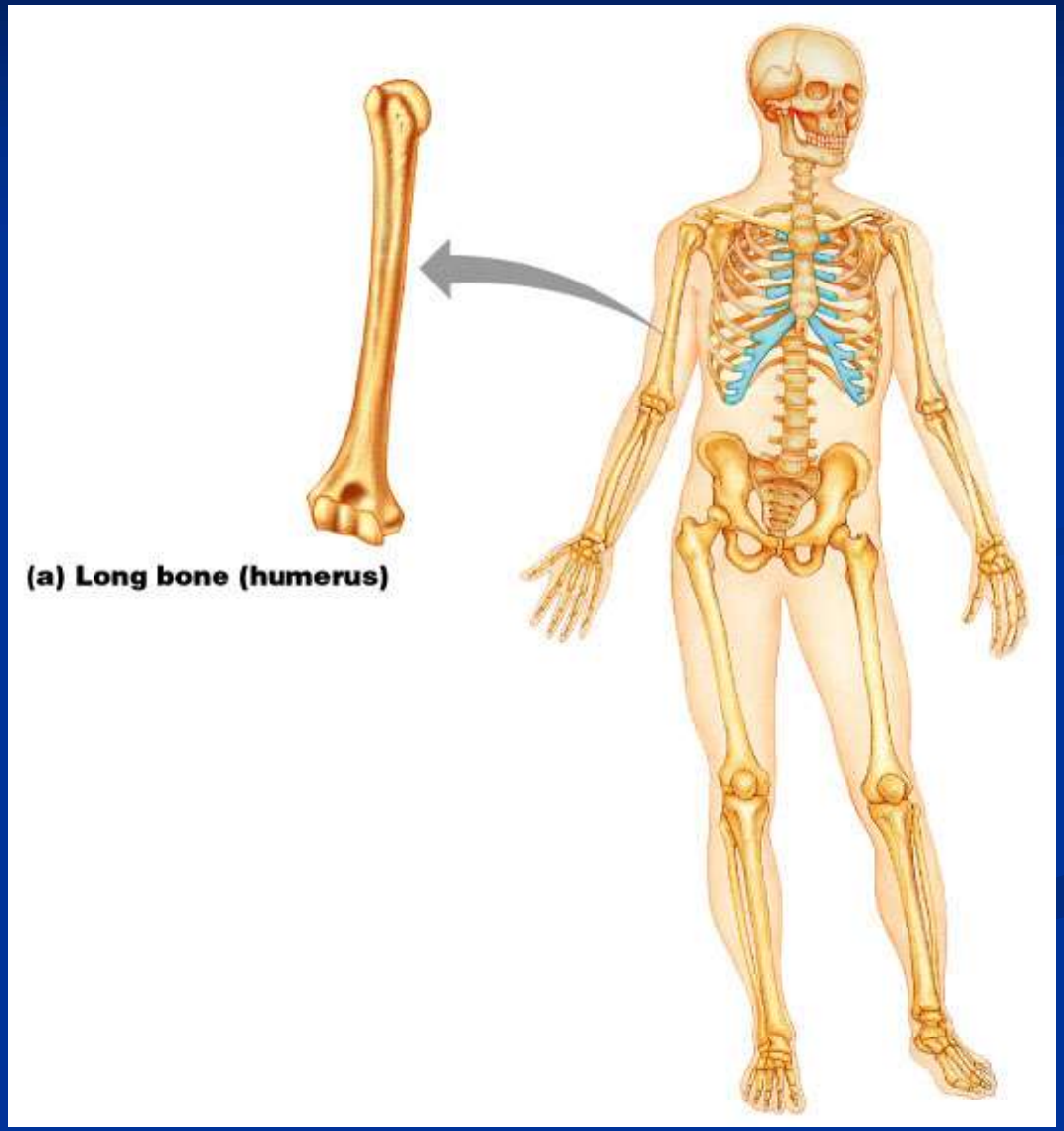
- Highly compressed with great tensile strength
- Contains collagen fibers
- Found in:
  - menisci of the knee
  - intervertebral discs

Growth of Cartilage

- Appositional
  - cells in the perichondrium secrete matrix against the external face of existing cartilage
- Interstitial
  - lacunae-bound chondrocytes inside the cartilage divide and secrete new matrix, expanding the cartilage from within
- Calcification of cartilage occurs
  - During normal bone growth
  - During old age

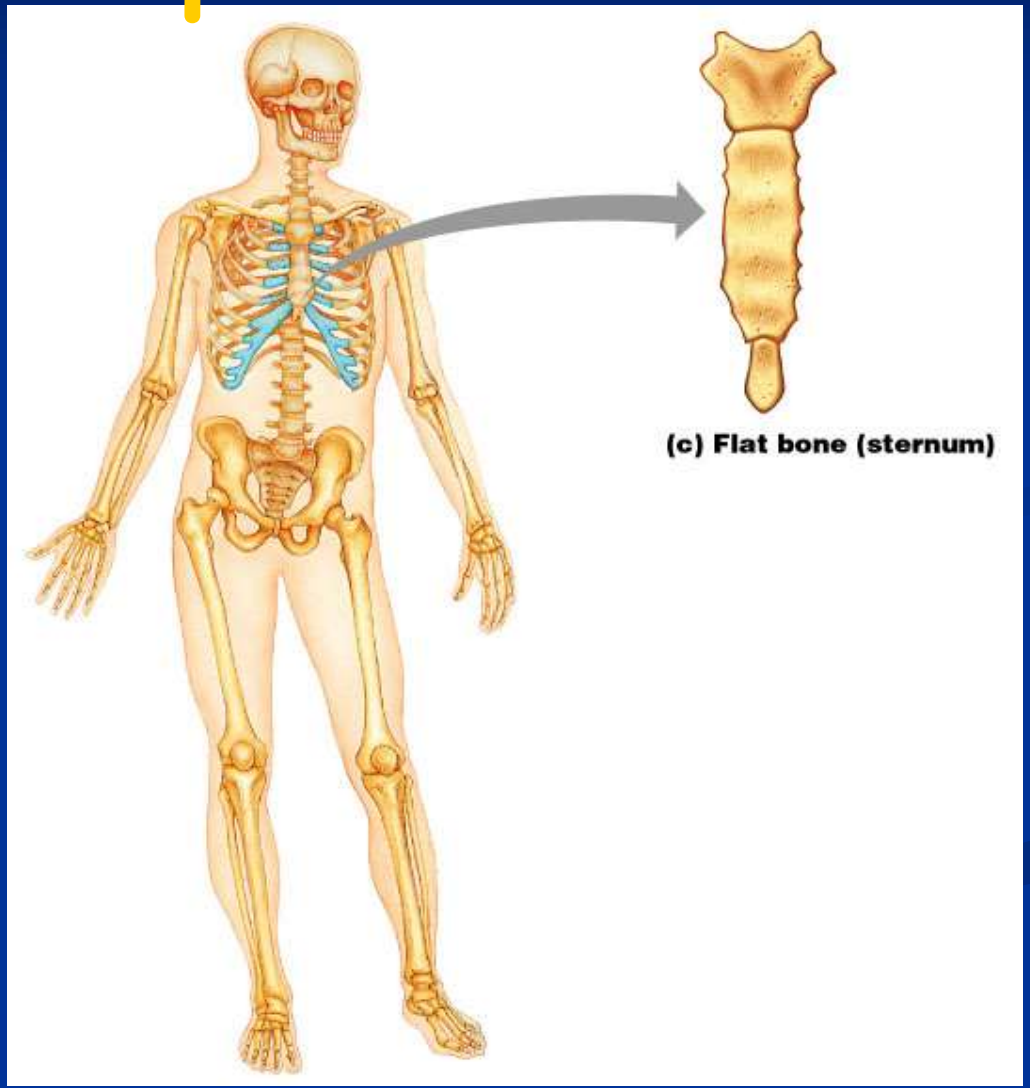
Classification of Bones: By Shape

- Long bones are longer than they are wide (e.g., humerus)



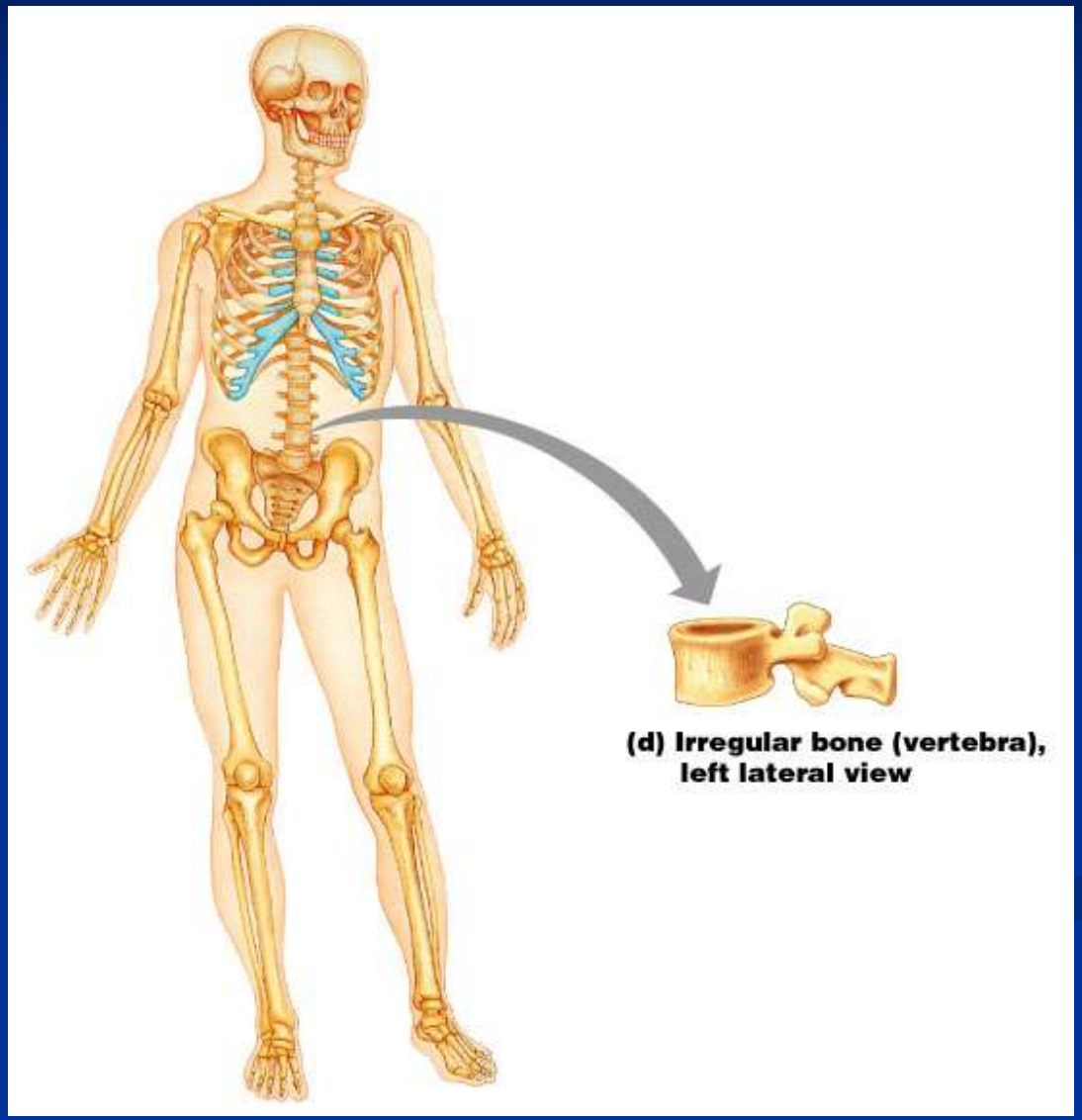
Classification of Bones: By Shape

- Flat bones are thin, flattened, and a bit curved (e.g., sternum, and most skull bones)



Classification of Bones: By Shape

- Irregular bones – bones with complicated shapes (e.g., vertebrae and hip bones)



Function of Bones

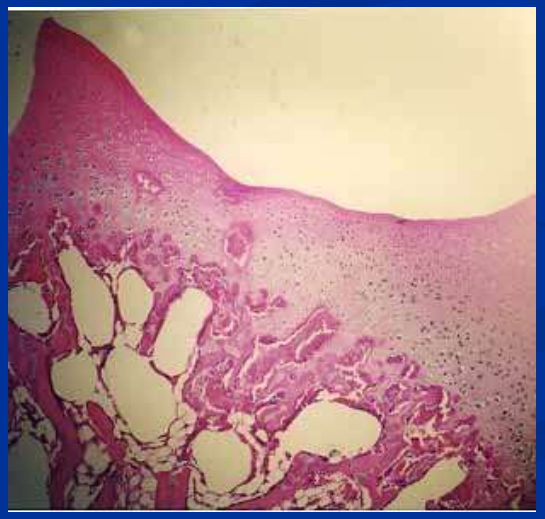
- Support
- form the framework that supports the body and cradles soft organs
- Protection – provide a protective case for the brain, spinal cord, and vital organs
- Movement – provide levers for muscles
- Mineral storage – reservoir for minerals, especially calcium and phosphorus
- Blood cell formation – hematopoiesis occurs within the marrow cavities of bones

Gross Anatomy of Bones: Bone Textures

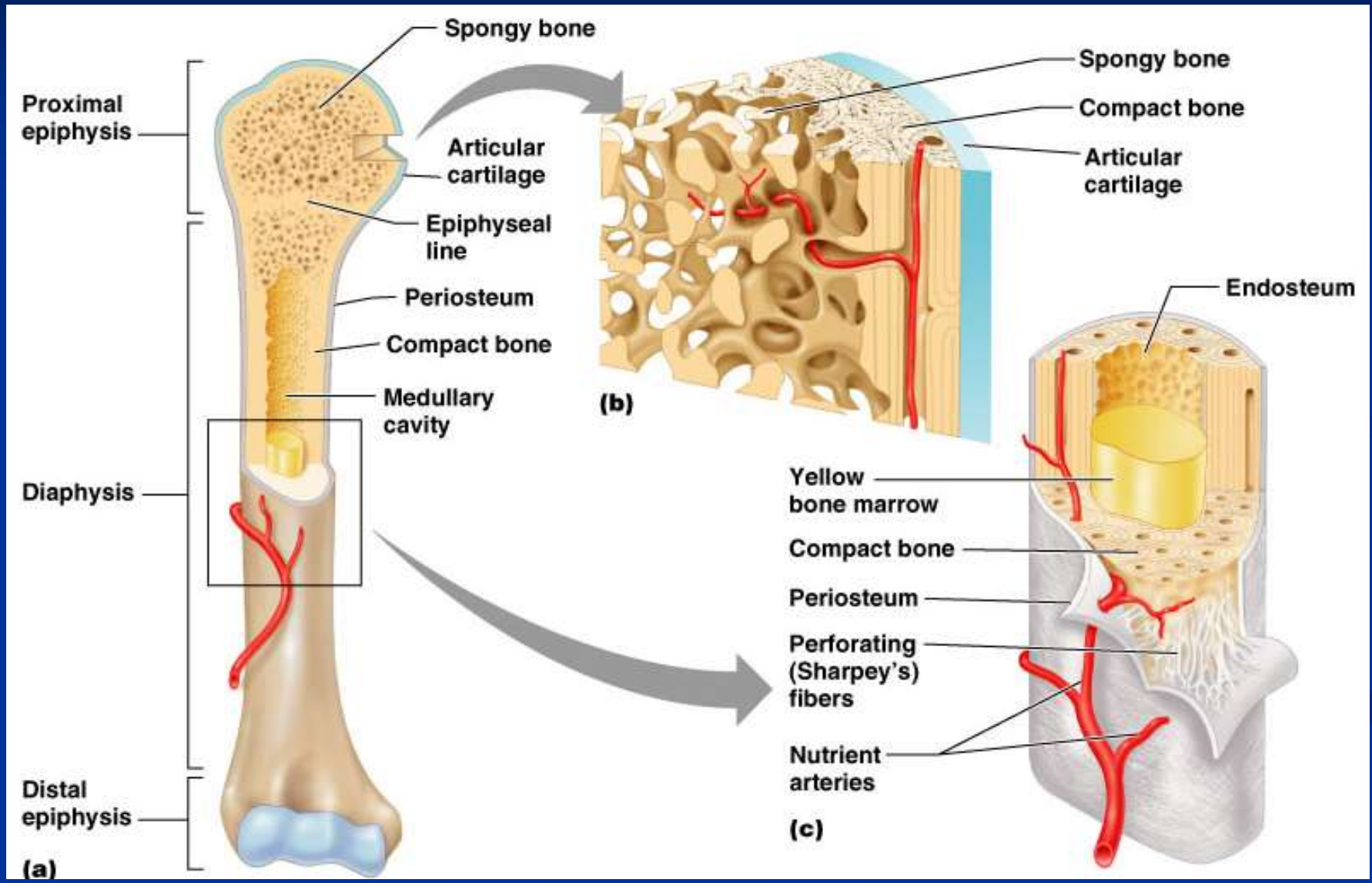
- Compact bone – dense outer layer



- Spongy bone – honeycomb of trabeculae filled with yellow bone marrow



Structure of Long Bone



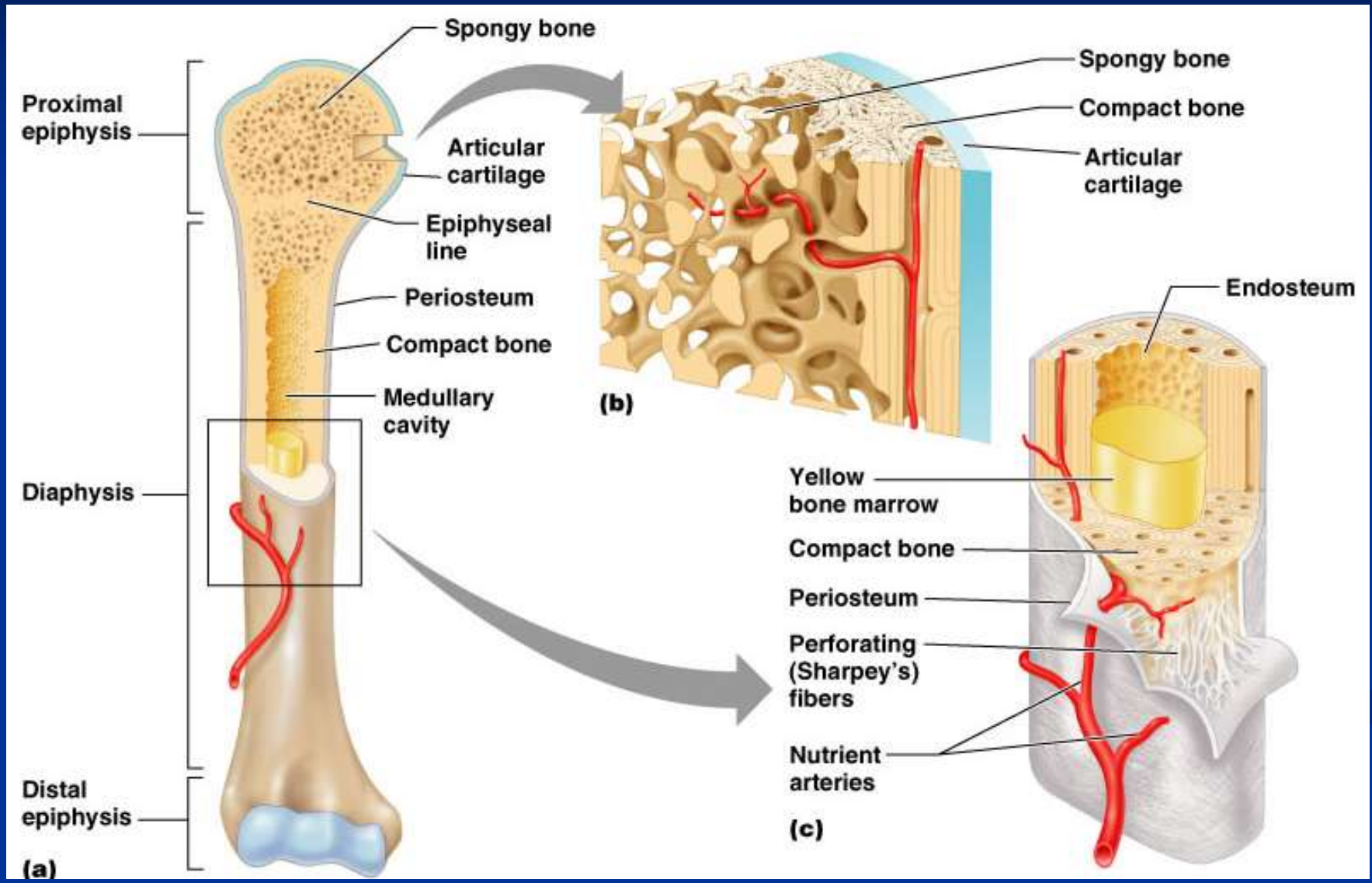
Structure of Long Bone

- Long bones consist of a diaphysis and an epiphysis
- Diaphysis
  - Tubular shaft that forms the axis of long bones
  - Composed of compact bone that surrounds the medullary cavity
  - Yellow bone marrow (fat) is contained in the medullary cavity

Structure of Long Bone

- Epiphyses
  - Expanded ends of long bones
  - Exterior is compact bone, and the interior is spongy bone
  - Joint surface is covered with articular (hyaline) cartilage
  - Epiphyseal line separates the diaphysis from the epiphyses

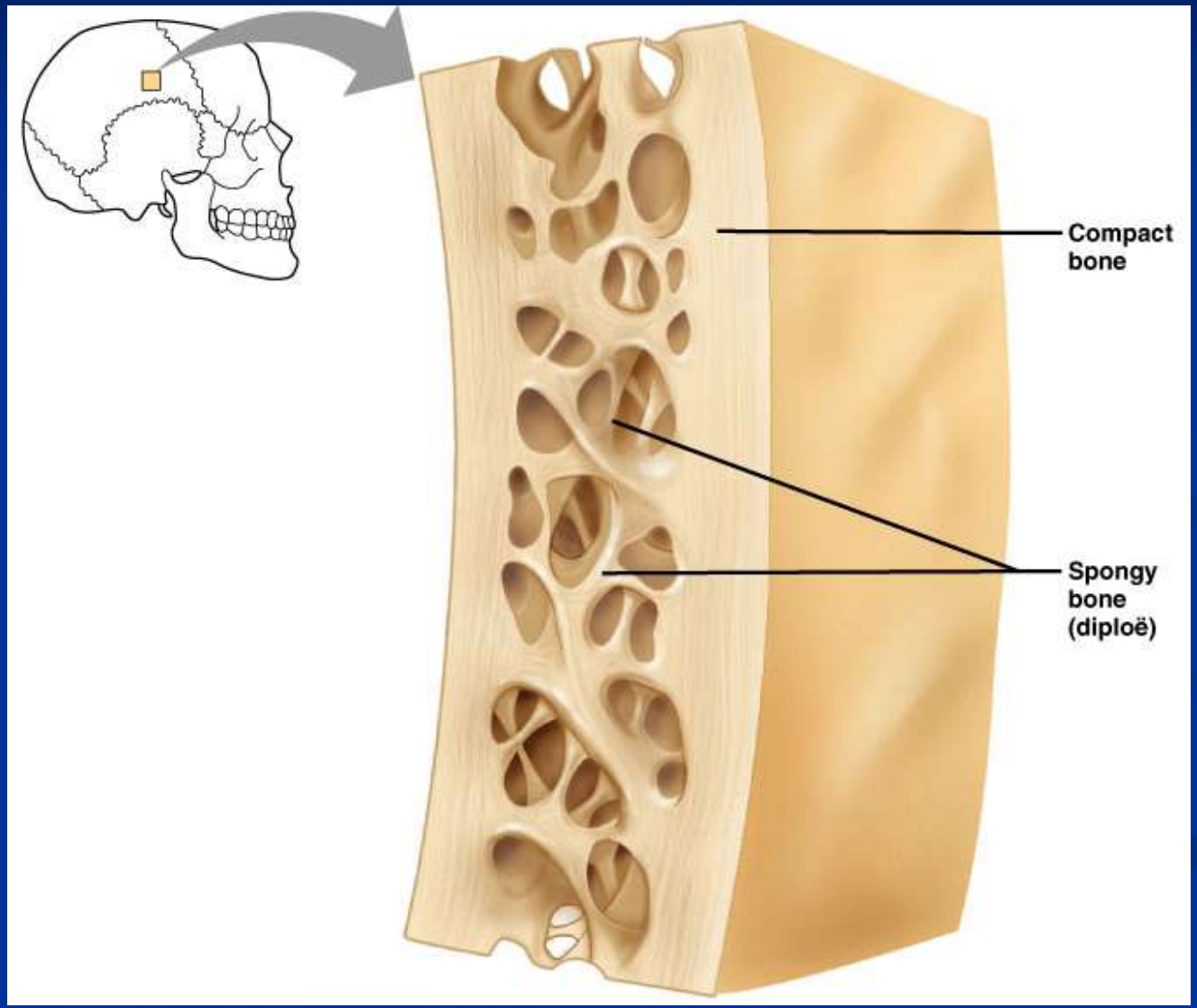
Structure of Long Bone



Bone Membranes

- Periosteum – double-layered protective membrane
  - Outer fibrous layer is dense regular connective tissue
  - Inner osteogenic layer is composed of osteoblasts and osteoclasts
  - Richly supplied with nerve fibers, blood, and lymphatic vessels, which enter the bone via nutrient foramina
  - Secured to underlying bone by Sharpey's fibers
- Endosteum – delicate membrane covering internal surfaces of bone

Structure of a Flat Bone



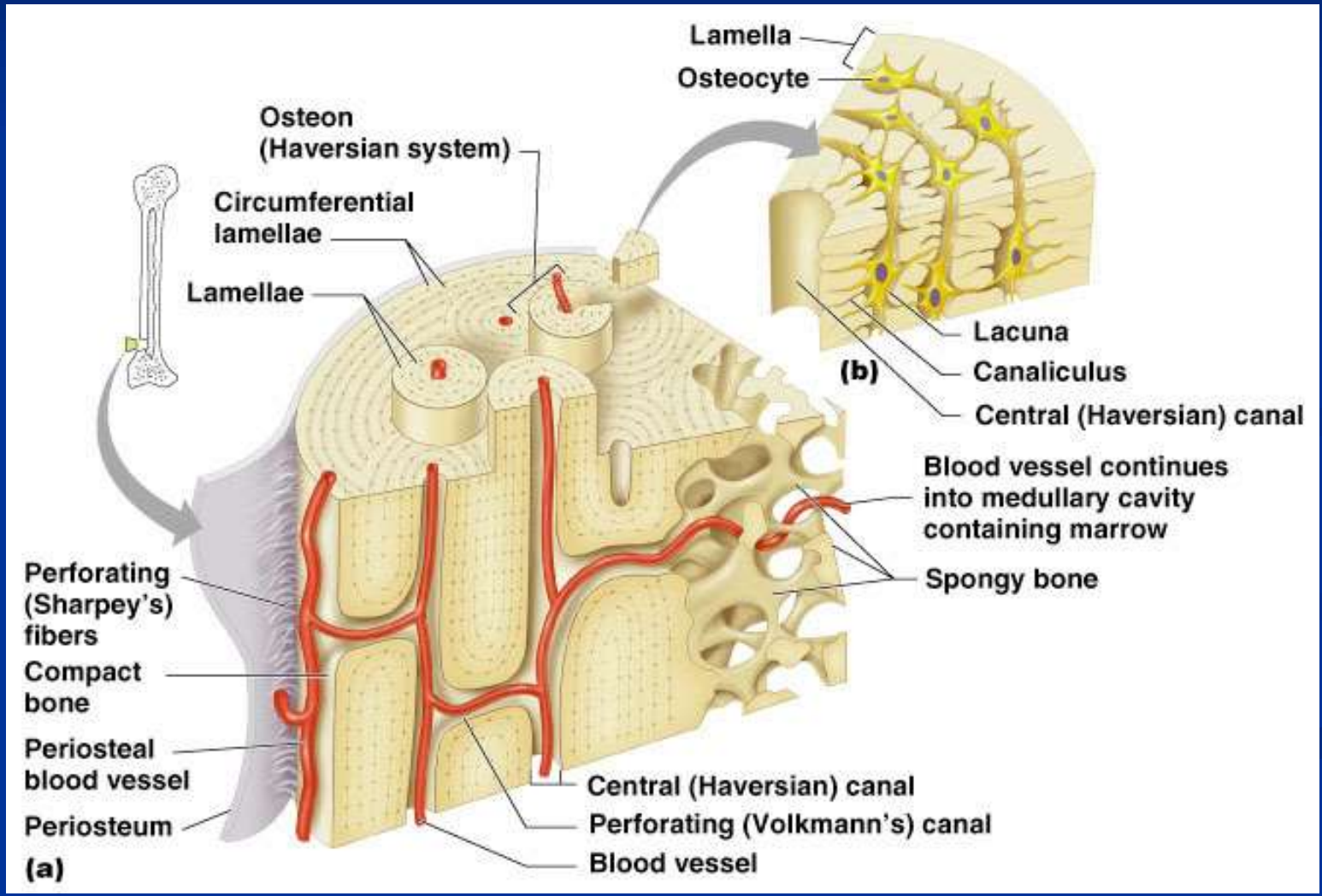
Structure of Short, Irregular, and Flat Bones

- Thin plates of periosteum-covered compact bone on the outside with endosteum-covered spongy bone (diploë) on the inside
- Have no diaphysis or epiphyses
- Contain bone marrow between the trabeculae

Location of Hematopoietic Tissue (Red Marrow)

- In infants
  - Found in the medullary cavity and all areas of spongy bone
- In adults
  - Found in the diploë of flat bones, and the head of the femur and humerus

Microscopic Structure of Bone: Compact Bone



Microscopic Structure of Bone: Compact Bone

- Haversian system, or osteon – the structural unit of compact bone
  - Lamella – weight-bearing, column-like matrix tubes composed mainly of collagen
  - Haversian, or central canal – central channel containing blood vessels and nerves
  - Volkmann's canals – channels lying at right angles to the central canal, connecting blood and nerve supply of the periosteum to that of the Haversian canal

Microscopic Structure of Bone: Compact Bone

- Osteocytes – mature bone cells
- Lacunae – small cavities in bone that contain osteocytes
- Canaliculi – hairlike canals that connect lacunae to each other and the central canal

Cells of Bone

- Osteoblasts – bone-forming cells
- Osteocytes – mature bone cells
- Osteoclasts – large cells that resorb or break down bone matrix

Chemical Composition of Bone: Organic

- Osteoid – unmineralized bone matrix composed of proteoglycans, glycoproteins, and collagen

Chemical Composition of Bone: Inorganic

- Hydroxyapatites, or mineral salts
  - Sixty-five percent of bone by mass
  - Mainly calcium phosphates
  - Responsible for bone hardness and its resistance to compression

Developmental Aspects of Bones

- Mesoderm gives rise to embryonic mesenchymal cells, which produce membranes and cartilages that form the embryonic skeleton
- The embryonic skeleton ossifies in a predictable timetable that allows fetal age to be easily determined from sonograms
- At birth, most long bones are well ossified (except for their epiphyses)

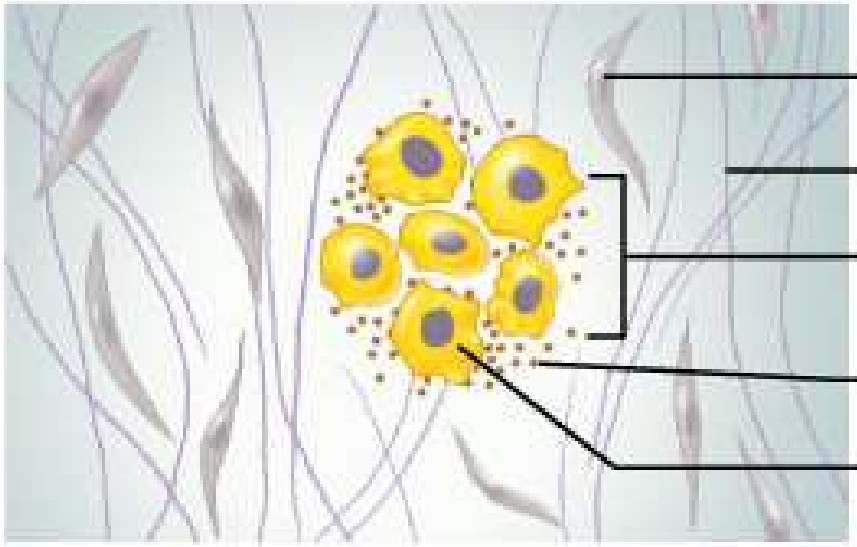
Developmental Aspects of Bones

- By age 25, nearly all bones are completely ossified
- In old age, bone resorption predominates
- A single gene that codes for vitamin D docking determines both the tendency to accumulate bone mass early in life, and the risk for osteoporosis later in life

Formation of Bone

- Intramembranous ossification – bone develops from a fibrous membrane
  - Formation of most of the flat bones of the skull and the clavicles
- Endochondral ossification – bone forms by replacing hyaline cartilage
  - Uses hyaline cartilage "bones" as models for bone construction
  - Requires breakdown of hyaline cartilage prior to ossification

Stages of Intramembranous Ossification

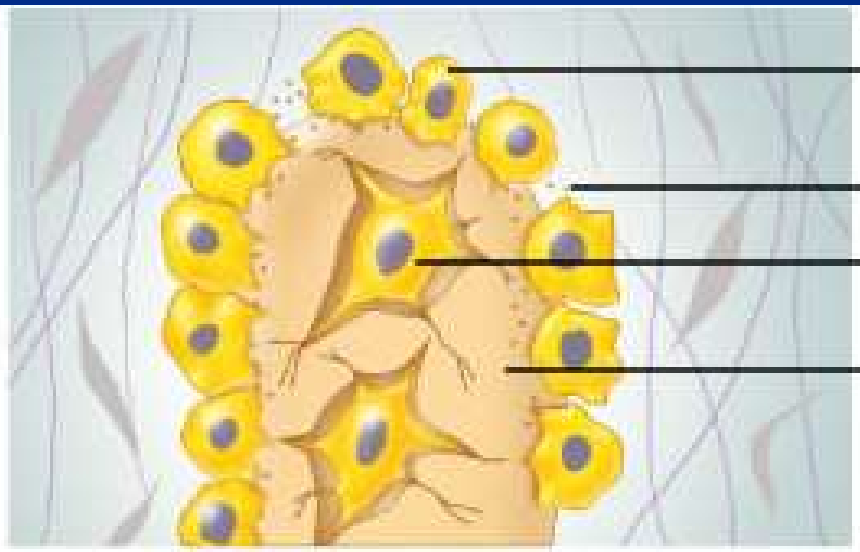


① **An ossification center appears in the fibrous connective tissue membrane.**

- Selected centrally located mesenchymal cells cluster and differentiate into osteoblasts, forming an ossification center.

Figure 6.7.1

Stages of Intramembranous Ossification

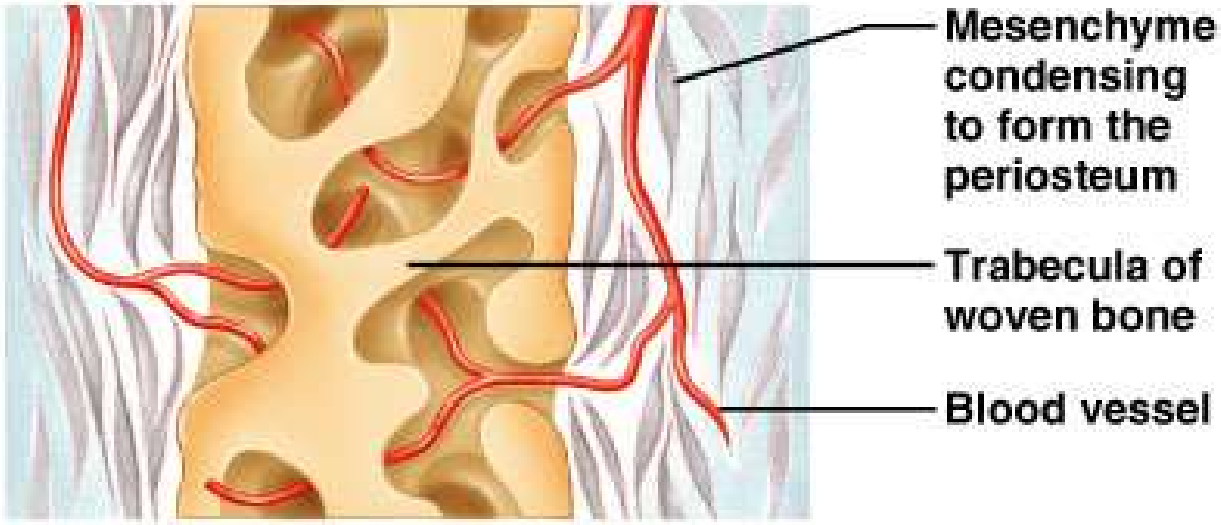


② **Bone matrix (osteoid) is secreted within the fibrous membrane.**

- Osteoblasts begin to secrete osteoid, which is mineralized within a few days.
- Trapped osteoblasts become osteocytes.

Figure 6.7.2

Stages of Intramembranous Ossification

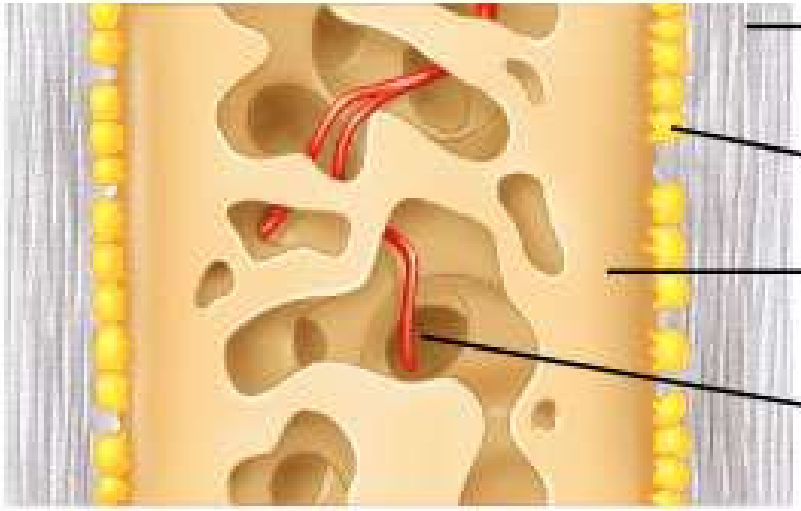


③ **Woven bone and periosteum form.**

- Accumulating osteoid is laid down between embryonic blood vessels, which form a random network. The result is a network (instead of lamellae) of trabeculae.
- Vascularized mesenchyme condenses on the external face of the woven bone and becomes the periosteum.

Figure 6.7.3

Stages of Intramembranous Ossification



④ **Bone collar of compact bone forms and red marrow appears.**

- Trabeculae just deep to the periosteum thicken, forming a woven bone collar that is later replaced with mature lamellar bone.
- Spongy bone (diploë), consisting of distinct trabeculae, persists internally and its vascular tissue becomes red marrow.

Figure 6.7.4

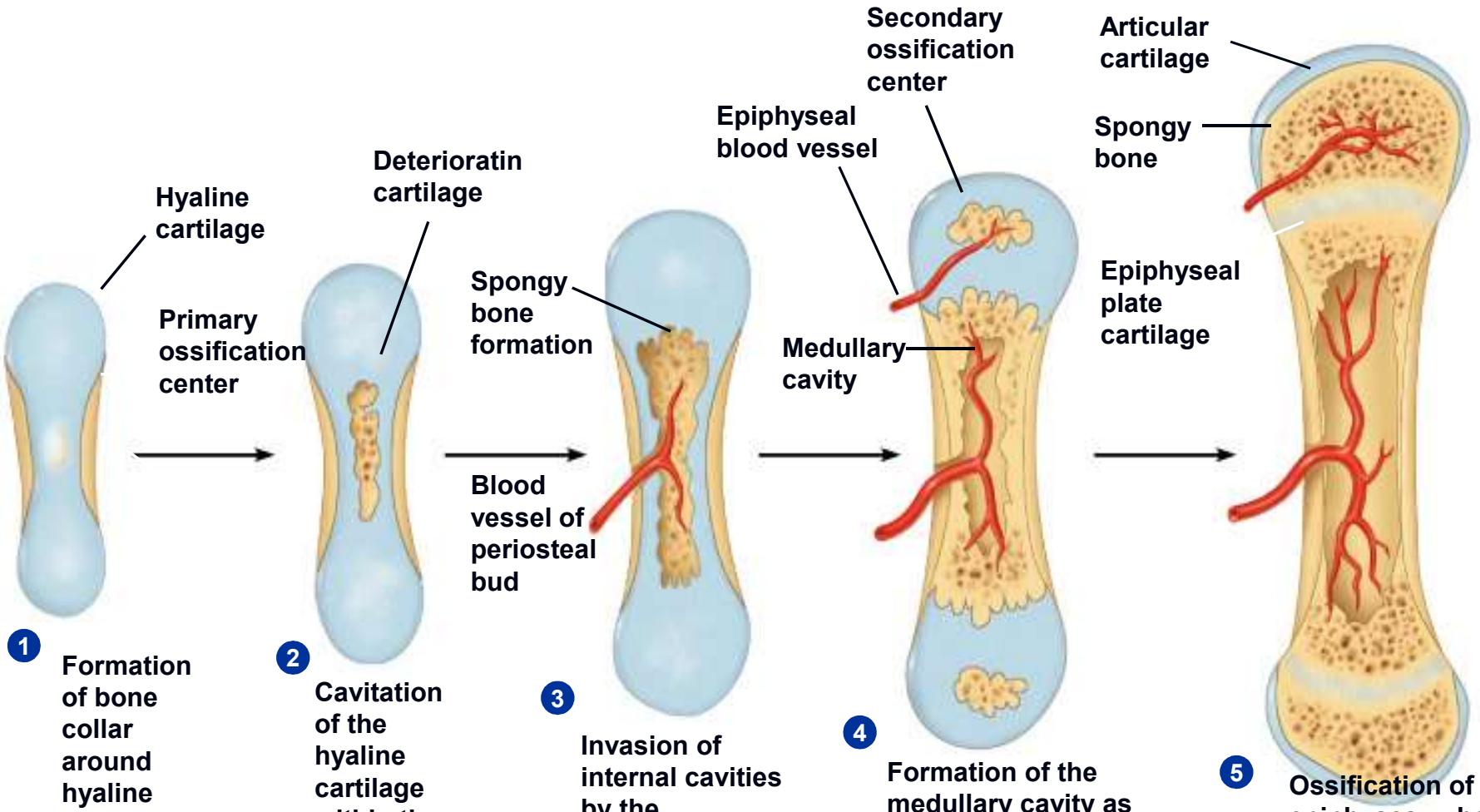
Stages of Intramembranous Ossification

- An ossification center appears in the fibrous connective tissue membrane
- Bone matrix is secreted within the fibrous membrane
- Woven bone and periosteum form
- Bone collar of compact bone forms, and red marrow appears

Endochondral Ossification

- Begins in the second month of development
- Uses hyaline cartilage "bones" as models for bone construction
- Requires breakdown of hyaline cartilage prior to ossification

Stages of Endochondral Ossification



Stages of Endochondral Ossification

- Formation of bone collar
- Cavitation of the hyaline cartilage
- Invasion of internal cavities by the periosteal bud, and spongy bone formation
- Formation of the medullary cavity; appearance of secondary ossification centers in the epiphyses
- Ossification of the epiphyses, with hyaline cartilage remaining only in the epiphyseal plates

Long Bone Growth and Remodeling

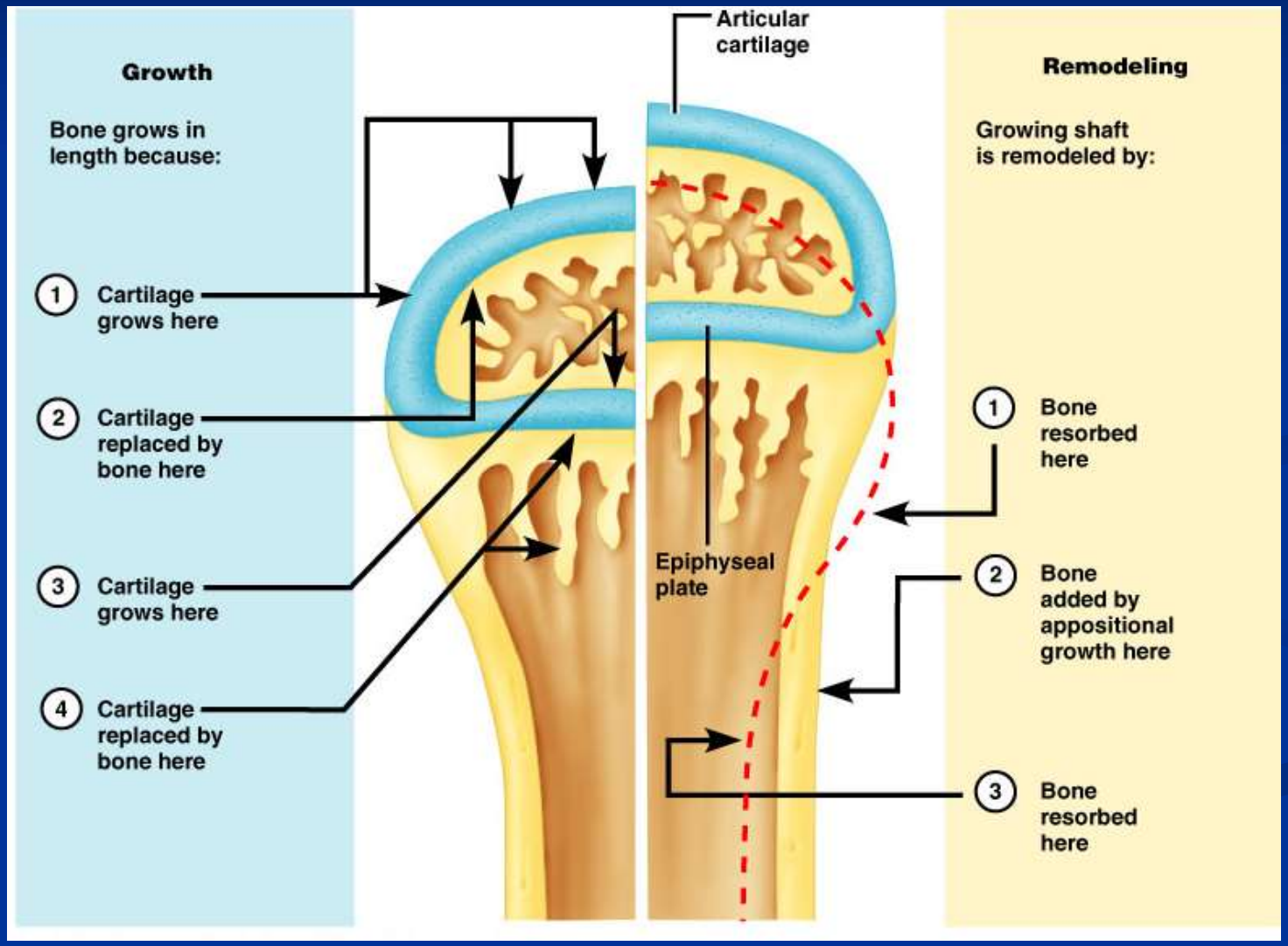


Figure 6.10

Postnatal Bone Growth

- Growth in length of long bones
  - Cartilage on the side of the epiphyseal plate closest to the epiphysis is relatively inactive
  - Cartilage abutting the shaft of the bone organizes into a pattern that allows fast, efficient growth
  - Cells of the epiphyseal plate proximal to the resting cartilage form three functionally different zones: growth, transformation, and osteogenic

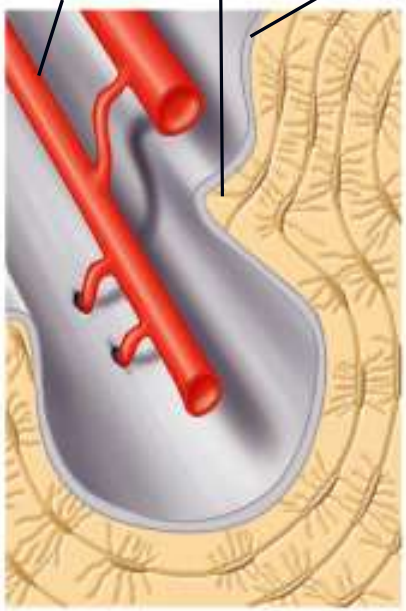
Functional Zones in Long Bone Growth

- Growth zone – cartilage cells undergo mitosis, pushing the epiphysis away from the diaphysis
- Transformation zone – older cells enlarge, the matrix becomes calcified, cartilage cells die, and the matrix begins to deteriorate
- Osteogenic zone – new bone formation occurs

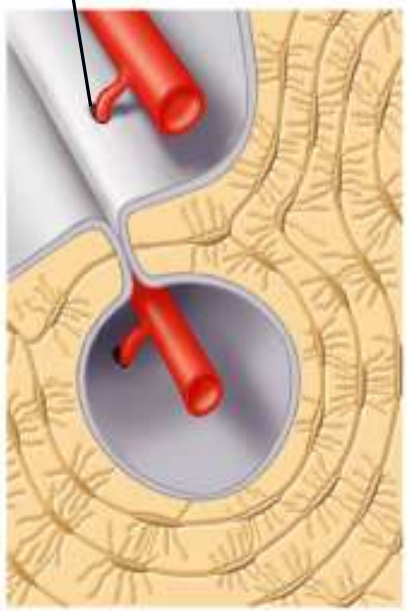
Long Bone Growth and Remodeling

- Growth in length – cartilage continually grows and is replaced by bone as shown
- Remodeling – bone is resorbed and added by appositional growth as shown in the next slide

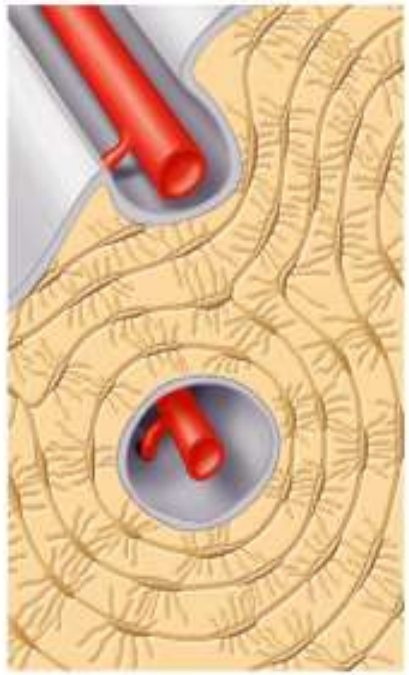
Appositional Growth of Bone



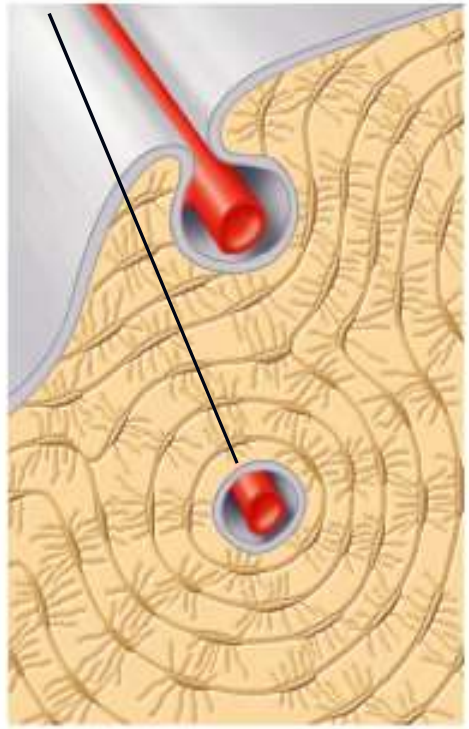
1. Osteoblasts beneath the periosteum secrete bone matrix, forming ridges that follow the course of periosteal blood vessels.



2. As the bony ridges enlarge and meet, the groove containing the blood vessel becomes a tunnel.



3. The periosteum lining the tunnel is transformed into an endosteum and the osteoblasts just deep to the tunnel endosteum secrete bone matrix, narrowing the canal.



4. As the osteoblasts beneath the endosteum form new lamellae, a new osteon is created. Meanwhile new circumferential lamellae are elaborated beneath the periosteum and the process is repeated, continuing to enlarge bone diameter.

Hormonal Regulation of Bone Growth During Youth

- During infancy and childhood, epiphyseal plate activity is stimulated by growth hormone
- During puberty, testosterone and estrogens:
  - Initially promote adolescent growth spurts
  - Cause masculinization and feminization of specific parts of the skeleton
  - Later induce epiphyseal plate closure, ending longitudinal bone growth

Bone Deposition

- Occurs where bone is injured or added strength is needed
- Requires a diet rich in protein, vitamins C, D, and A, calcium, phosphorus, magnesium, and manganese
- Alkaline phosphatase is essential for mineralization of bone
- Sites of new matrix deposition are revealed by the:
  - Osteoid seam – unmineralized band of bone matrix
  - Calcification front – abrupt transition zone between the osteoid seam and the older mineralized bone

Bone Resorption

- Accomplished by osteoclasts
- Resorption bays – grooves formed by osteoclasts as they break down bone matrix
- Resorption involves osteoclast secretion of:
  - Lysosomal enzymes that digest organic matrix
  - Acids that convert calcium salts into soluble forms
- Dissolved matrix is transcytosed across the osteoclast's cell where it is secreted into the interstitial fluid and then into the blood

Importance of Ionic Calcium in the Body

- Calcium is necessary for:
  - Transmission of nerve impulses
  - Muscle contraction
  - Blood coagulation
  - Secretion by glands and nerve cells
  - Cell division

Control of Remodeling

- Two control loops regulate bone remodeling
  - Hormonal mechanism maintains calcium homeostasis in the blood
  - Mechanical and gravitational forces acting on the skeleton

Hormonal Mechanism

- Rising blood Ca^{2+} levels trigger the thyroid to release calcitonin
- Calcitonin stimulates calcium salt deposit in bone
- Falling blood Ca^{2+} levels signal the parathyroid glands to release PTH
- PTH signals osteoclasts to degrade bone matrix and release Ca^{2+} into the blood

Hormonal Mechanism

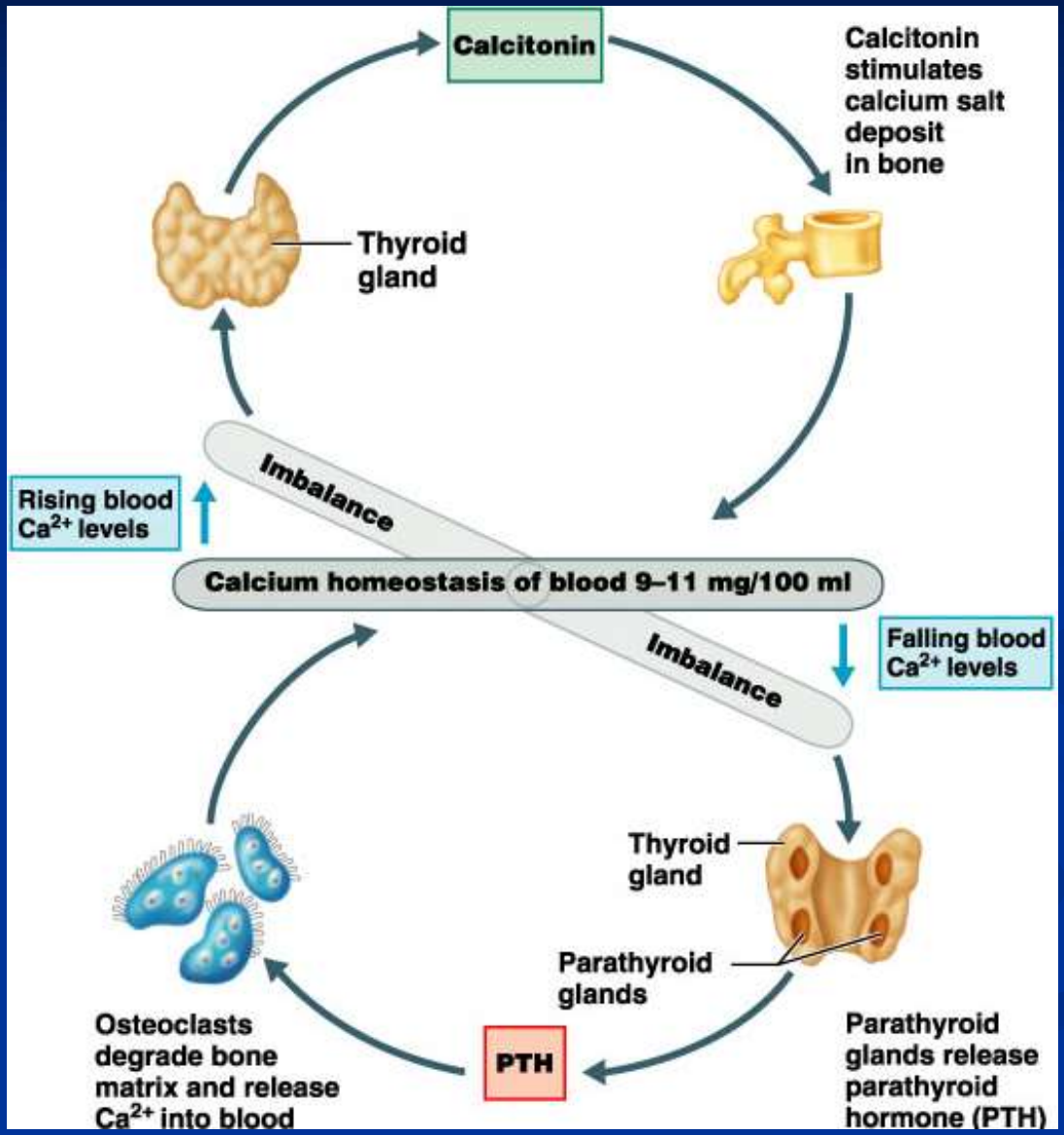


Figure 6.12

Response to Mechanical Stress

- Wolff's law – a bone grows or remodels in response to the forces or demands placed upon it
- Observations supporting Wolff's law include
  - Long bones are thickest midway along the shaft (where bending stress is greatest)
  - Curved bones are thickest where they are most likely to buckle

Response to Mechanical Stress

- Trabeculae form along lines of stress
- Large, bony projections occur where heavy, active muscles attach

Response to Mechanical

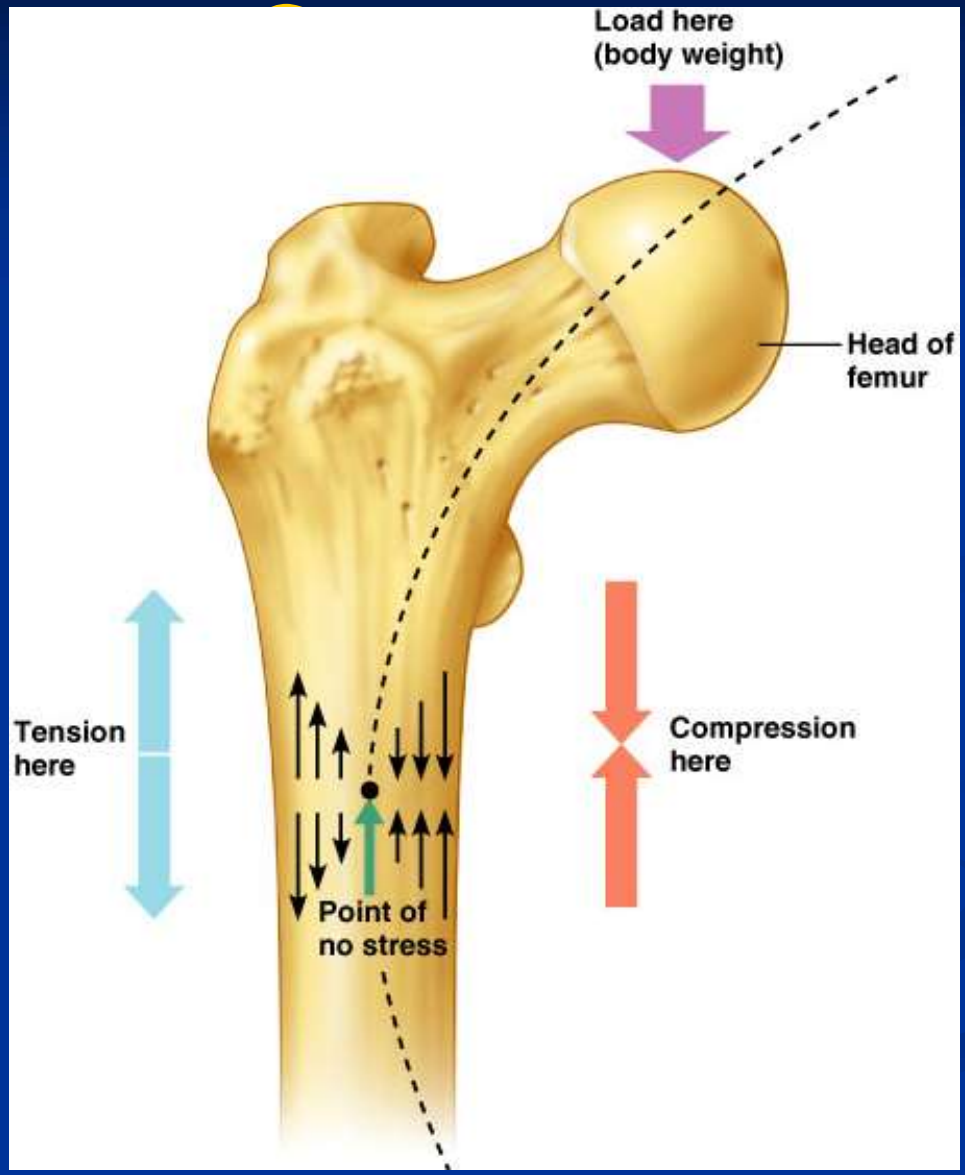


Figure 6.13

Bone Fractures (Breaks)

- Classified by:
  - The position of the bone ends after fracture
  - The completeness of the break
  - The orientation of the bone to the long axis
  - Whether or not the bones ends penetrate the skin

Types of Bone Fractures

- Nondisplaced – bone ends retain their normal position
- Displaced – bone ends are out of normal alignment
- Complete – bone is broken all the way through
- Incomplete – bone is not broken all the way through
- Linear – the fracture is parallel to the long axis of the bone

Types of Bone Fractures

- Transverse – the fracture is perpendicular to the long axis of the bone
- Compound (open) – bone ends penetrate the skin
- Simple (closed) – bone ends do not penetrate the skin

Common Types of Fractures

- Comminuted – bone fragments into three or more pieces; common in the elderly
- Spiral – ragged break when bone is excessively twisted; common sports injury
- Depressed – broken bone portion pressed inward; typical skull fracture
- Compression – bone is crushed; common in porous bones

Common Types of Fractures

- Epiphyseal – epiphysis separates from diaphysis along epiphyseal line; occurs where cartilage cells are dying
- Greenstick – incomplete fracture where one side of the bone breaks and the other side bends; common in children

Common Types of Fractures

TABLE 6.2 Common Types of Fractures

Fracture Type	Description and Comments	Fracture Type	Description and Comments
Comminuted	Bone fragments into three or more pieces Particularly common in the aged, whose bones are more brittle	Compression	Bone is crushed Common in porous bones (i.e., osteoporotic bones) subjected to extreme trauma, as in a fall



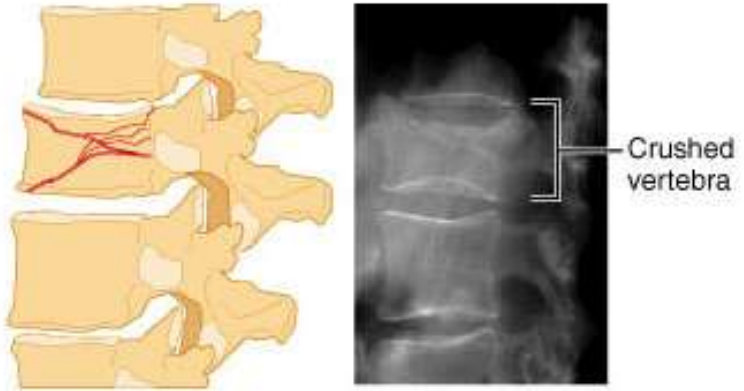
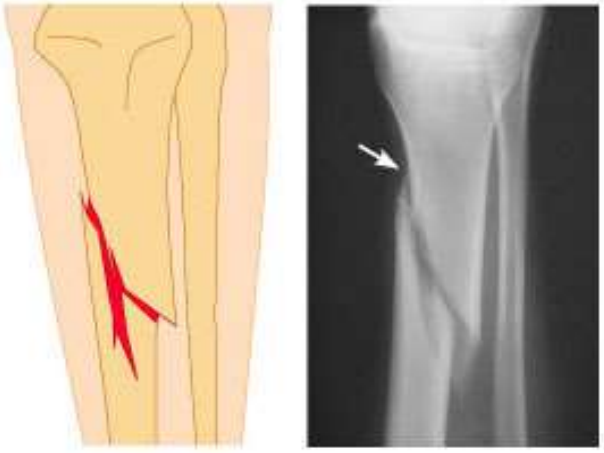


Table 6.2.1

Common Types of Fractures

TABLE 6.2 Common Types of Fractures *(continued)*

Fracture Type	*Description and Comments*	*Fracture Type*	*Description and Comments*
Spiral	Ragged break occurs when excessive twisting forces are applied to a bone Common sports fracture	Epiphyseal	Epiphysis separates from the diaphysis along the epiphyseal plate Tends to occur where cartilage cells are dying and calcification of the matrix is occurring



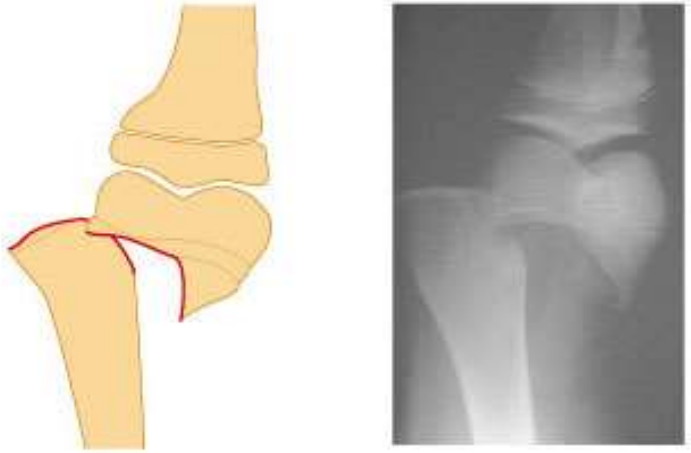


Table 6.2.2

Common Types of Fractures

TABLE 6.2 Common Types of Fractures *(continued)*

Fracture Type	*Description and Comments*	*Fracture Type*	*Description and Comments*
Depressed	Broken bone portion is pressed inward Typical of skull fracture	Greenstick	Bone breaks incompletely, much in the way a green twig breaks. Only one side of the shaft breaks; the other side bends Common in children, whose bones have relatively more organic matrix and are more flexible than those of adults



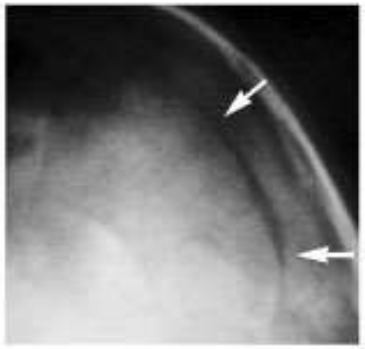






Table 6.2.3

Stages in the Healing of a Bone Fracture

- Hematoma formation
  - Torn blood vessels hemorrhage
  - A mass of clotted blood (hematoma) forms at the fracture site
  - Site becomes swollen, painful, and inflamed

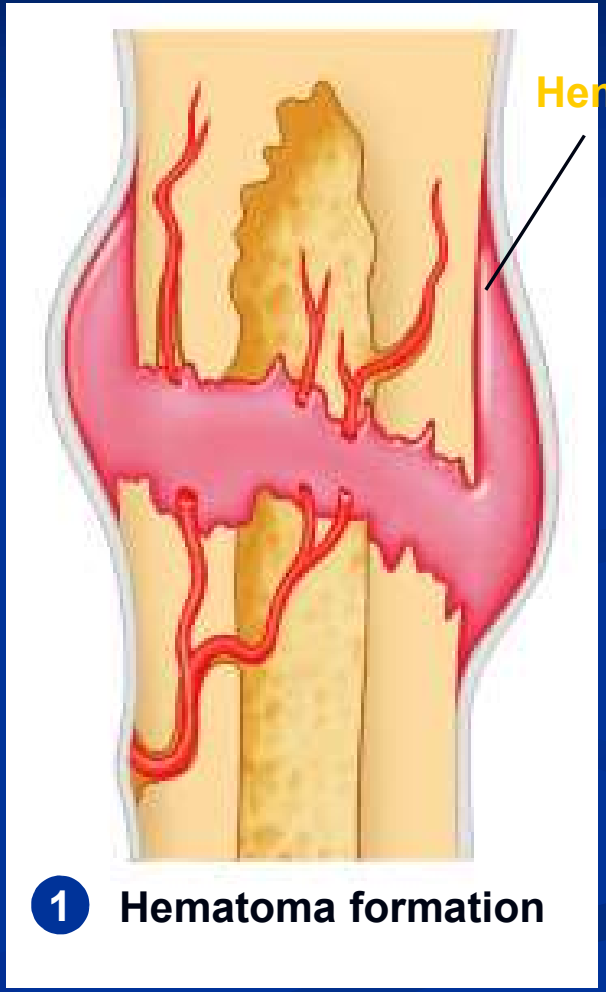


Figure 6.14.1

Stages in the Healing of a Bone Fracture

- Fibrocartilaginous callus forms
- Granulation tissue (soft callus) forms a few days after the fracture
- Capillaries grow into the tissue and phagocytic cells begin cleaning debris

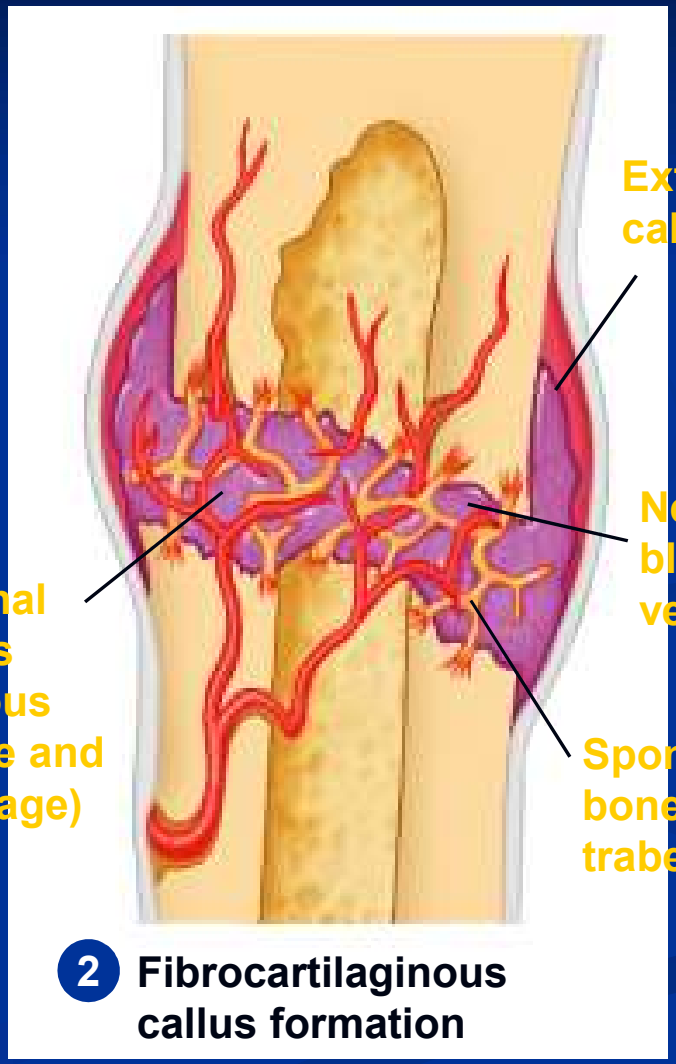


Figure 6.14.2

Stages in the Healing of a Bone Fracture

- Bony callus formation
  - New bone trabeculae appear in the fibrocartilaginous callus
  - Fibrocartilaginous callus converts into a bony (hard) callus
  - Bone callus begins 3-4 weeks after injury, and continues until firm union is formed 2-3 months later



Figure 6.14.3

Stages in the Healing of a Bone Fracture

- Bone remodeling
  - Excess material on the bone shaft exterior and in the medullary canal is removed
  - Compact bone is laid down to reconstruct shaft walls



Figure 6.14.4

Stages in the Healing of a Bone Fracture

- The fibrocartilaginous callus forms when:
  - Osteoblasts and fibroblasts migrate to the fracture and begin reconstructing the bone
  - Fibroblasts secrete collagen fibers that connect broken bone ends
  - Osteoblasts begin forming spongy bone
  - Osteoblasts furthest from capillaries secrete an externally bulging cartilaginous matrix that later calcifies

Homeostatic Imbalances

- Rickets
  - Bones of children are inadequately mineralized causing softened, weakened bones
  - Bowed legs and deformities of the pelvis, skull, and rib cage are common
  - Caused by insufficient calcium in the diet, or by vitamin D deficiency

Homeostatic Imbalances

- Osteoporosis
  - Group of diseases in which bone reabsorption outpaces bone deposit
  - Spongy bone of the spine is most vulnerable
  - Occurs most often in postmenopausal women
  - Bones become so fragile that sneezing or stepping off a curb can cause fractures